

VitalFlo

Understanding Spirometry Curves

Key Definitions

Vocabulary to navigate the table above, in addition to the values noted on the curves (left).

FEV₁ **Forced Exhalatory Volume 1 (Liters)**
Volume measured in the first second of the spirometry blow.

FVC **Forced Vital Capacity (Liters)**
Maximum volume of gas that a patient can exhale after maximal inhale.

FEV₁/FVC **Forced Exhalatory Volume 1 to Forced Vital Capacity Ratio**
Proportion of FEV₁ to FVC shown as a percentage, or how much one can exhale in the first second compared to their full ability to exhale maximally.

FEF₂₅₋₇₅ **Forced Exhalatory Flow (L/sec) measured from 25% to 75%**
The flow rate (change in volume over change in time) for the middle range of the spirometry blow.

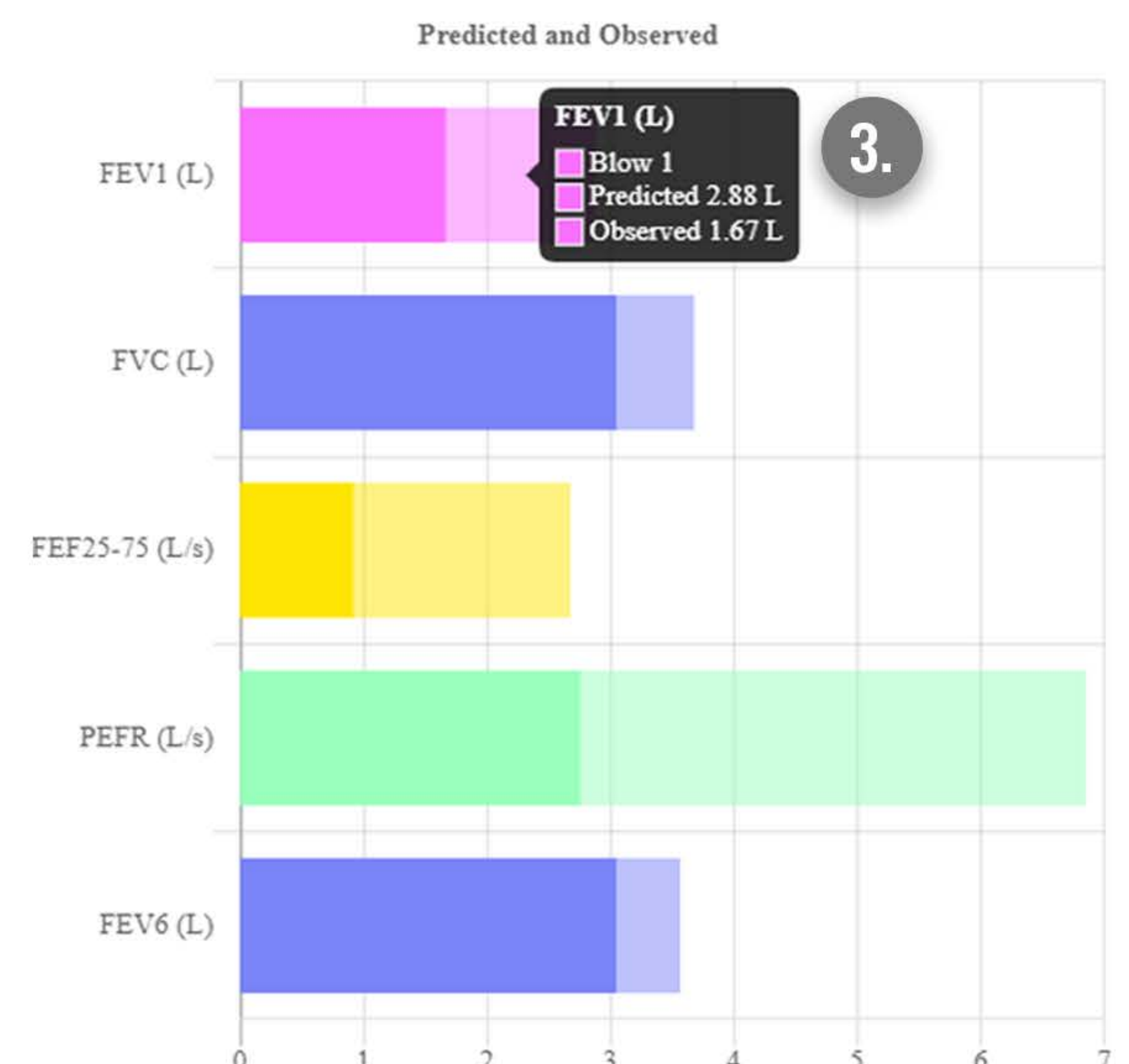
PEFR **Peak Exhalatory Flow Rate (L/sec)**
The maximum flow rate (change in volume over change in time).

FEV₆ **Forced Exhalatory Volume 6 (L)**
Volume measured in the final (sixth) second of the spirometry blow.

1. Blows toggle between the blows.
2. See differences between **observed values**, **percent predicted** and the **predicted values**.
3. Hover for values.

1. [Blow 1](#) Blow 2 Blow 3

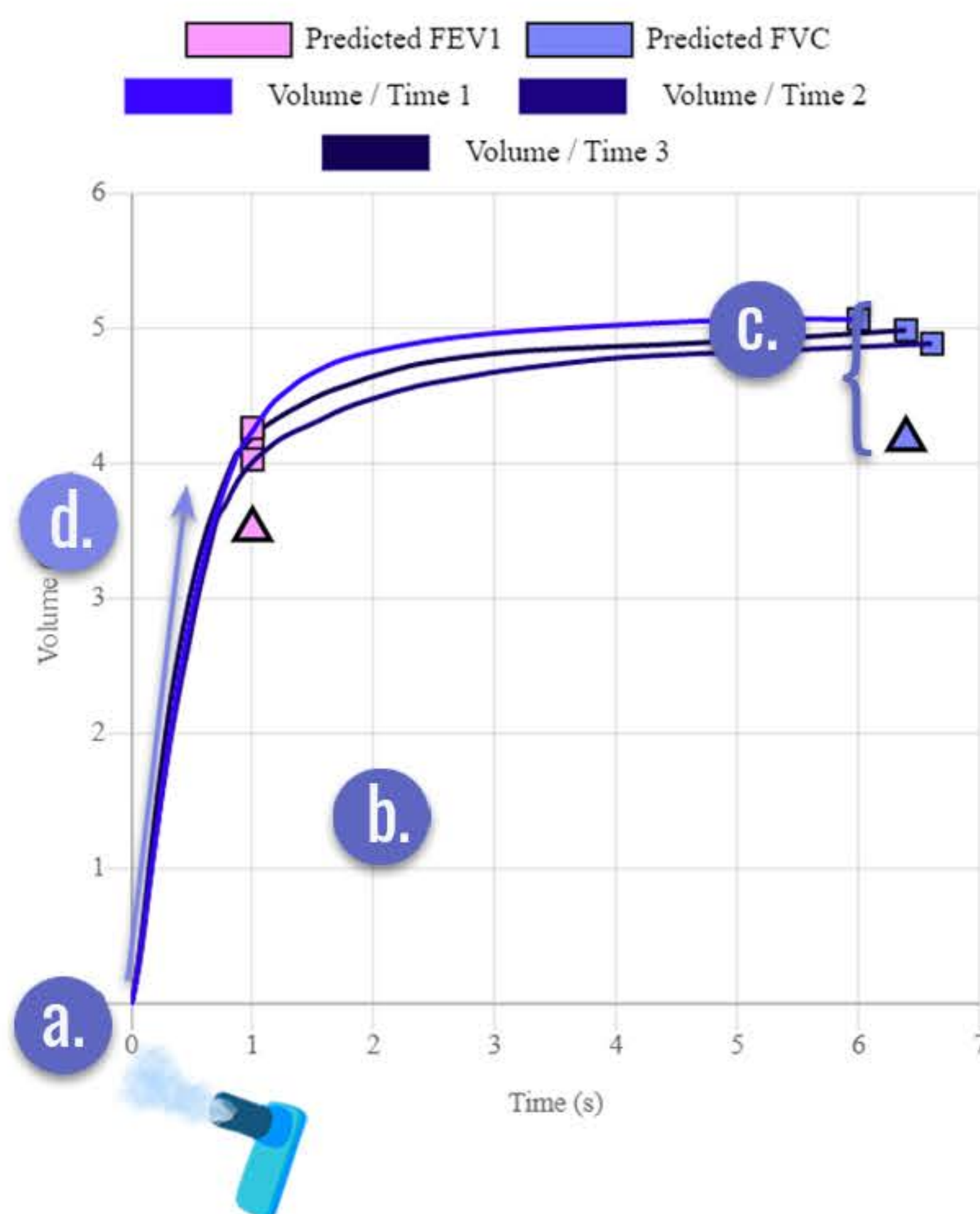
1	2.	Observed	Percentage	Predicted
 FEV ₁ (L)	1.67	58.0	2.88	
 FVC (L)	3.05	82.9	3.68	
 FEV ₁ /FVC	54.8	70.1	78.2	
 FEF ₂₅₋₇₅ (L/s)	0.92	34.4	2.68	
 PEFR (L/s)	2.76	40.3	6.85	
 FEV ₆ (L)	3.05	85.5	3.57	



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Flow / Volume Volume / Time Flow / Time

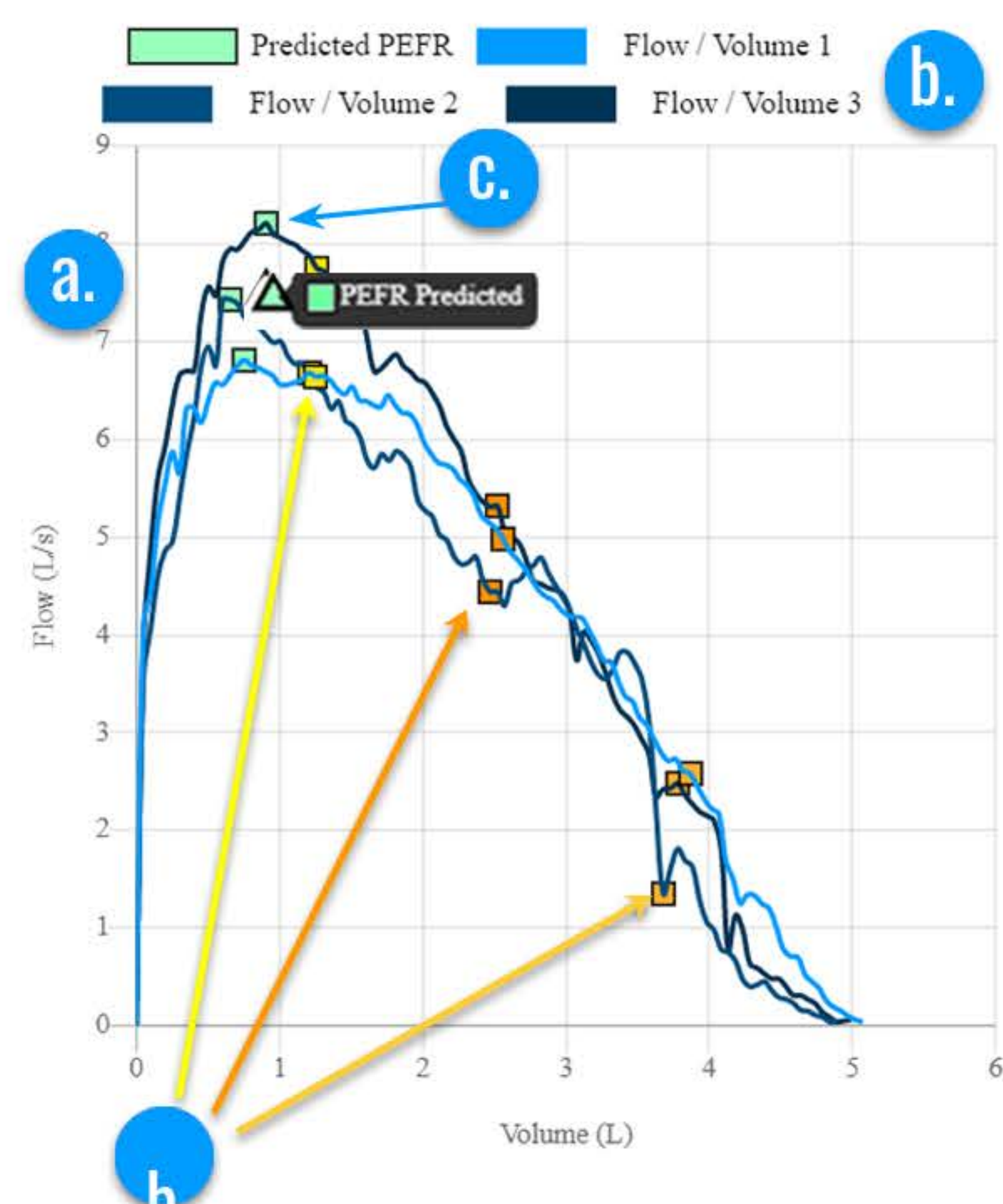


Volume-Time Curve

Exhale of blow represented visually over time: air volume in liters exhaled over seconds (we request 6 seconds)

- a. Patient inhales before this curve, then exhales maximally into the spirometer (start of curve).
- b. ■ Squares represent actual values recorded by spirometer. Hovering shows which blow.
- c. ▲ Triangle represents predicted values by selected predicted equation (e.g. GLI).
You can then look at distance between predicted and actual.
- d. ↑ While not needed here, this slope equates to PEFR in other curves, or the steepest rate of volume/time.

Flow / Volume Volume / Time Flow / Time

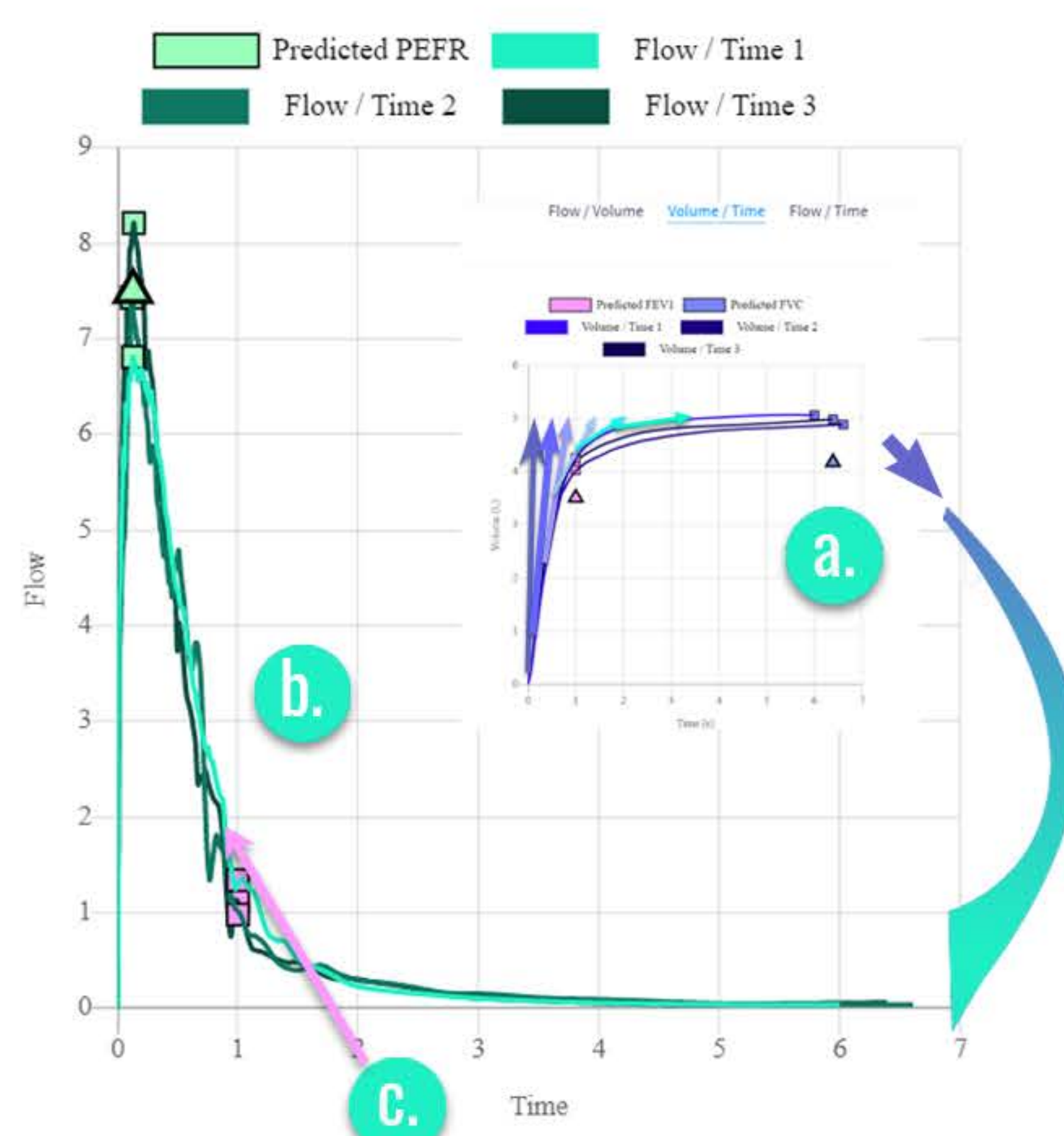


Flow-Volume Curve

Flow refers to change in volume over change in time and can be represented by the slope in d. above. The highest point on this curve would be the max flow rate, or the PEFR.

- a. Hovering over triangles reveal the predicted value.
- b. The legend at the top represents the colors on the graph for lines e.g. — Flow / Volume 1 refers to the blue middle line on the curve (first blow).
- c. PEFR represents maximum flow, then as you exhale, flow will decrease as the volume expelled from lungs increases.
- d. These squares represent points at ■ 25%, ■ 50%, and ■ 75% of FEF (the middle half of the maneuver).

Flow / Volume Volume / Time Flow / Time



Flow-Time Curve

Represents flow (slope of volume over time) against time in seconds.

- a. This curve can be thought of as the series of slopes seen in the Volume-Time Curve (e.g. PEFR was one) mapping flow, against length of the blow (time in seconds).
- b. Here, you can again see that PEFR is the at the peak of flow (y-axis).
- c. This curve also denotes ■FEV1, lined up with the first second on the x-axis.